MANIPAL
ACADEMY of HIGHER EDUCATION
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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Machine Learning-Based Prediction And Recommendation Of Roommate Compatibility Using Lifestyle Preferences

Project Report

Applied Probability and Statistics Lab (AME 5152)

submitted to

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Abstract

Selecting a suitable roommate is a multi-attribute decision problem because individuals differ in their daily routines, lifestyle habits, financial requirements, and expectations about shared living spaces.

This project investigates the application of machine learning classification and statistical techniques to predict roommate compatibility. The experiment uses the training_data.csv dataset, containing 1,333 records and 31 columns. The compatibility score is converted into a binary target variable using a threshold of 70, where scores greater than or equal to 70 represent Compatible and scores below 70 represent Not Compatible.

Logistic Regression is used as the baseline classifier, while Random Forest is evaluated as the improved classifier. Both models are trained and evaluated using the same held-out test set. The experiment incorporates stratified group sampling, paired bootstrap resampling, confidence interval estimation, and hypothesis testing using the exact McNemar test.

Random Forest achieved an accuracy of 71.16%, compared with 62.92% for Logistic Regression. The observed improvement was 8.24 percentage points. The 95% bootstrap confidence interval ranged from 2.62% to 13.87%, while the exact one-sided McNemar test produced a p-value of 0.003574.

The statistical results provide evidence that Random Forest performs better than Logistic Regression on the evaluated test set. The experiment demonstrates how probability and statistical methods can be applied to assess model performance and determine whether an observed improvement is statistically meaningful.

1. Problem Statement and Dataset

1.1 Problem Statement

Finding a suitable roommate can be challenging because individuals have different lifestyles, daily habits, personal preferences, and financial requirements. Differences in sleeping schedules, cleanliness expectations, noise tolerance, cooking habits, guest preferences, smoking preferences, and financial arrangements can create disagreements in shared accommodation. Traditional roommate selection often relies on limited information or informal discussions and may not systematically consider several factors together.

This project aims to develop a machine learning classification approach that predicts roommate compatibility based on the lifestyle preferences, personal characteristics, and financial requirements of two individuals. The project compares a baseline classification model with an improved model and applies statistical techniques to determine whether the observed performance improvement is meaningful and reliable.

1.2 Project Objectives

1. To inspect and preprocess the roommate compatibility dataset.
2. To identify relevant numerical and categorical features for classification.
3. To convert the compatibility score into a binary classification target.
4. To implement Logistic Regression as the baseline model and Random Forest as the improved model.
5. To evaluate both models using the same held-out test dataset.
6. To apply bootstrap sampling, confidence interval estimation, and hypothesis testing to assess the reliability of model improvement.

1.3 Dataset Description

The dataset used in this project is the uploaded file training_data.csv. It contains 1,333 records and 31 columns. Each record represents a pair of students with their respective personal details, lifestyle preferences, financial requirements, and a compatibility score.

The dataset contains information about both individuals, including their budget range, university, course, year of study, age, nationality, property preferences, smoking preferences, pet preferences, cleanliness, and noise tolerance.

The compatibility score is used to construct a binary classification target for evaluating machine learning models.

Dataset property	**Value**
Dataset file	training_data.csv
Number of profiles	1,333
Number of columns	31
Target column	compatibility_score
Missing values	0
Data type	Numerical and categorical
Classification type	Binary classification

1.4 Attribute Categories

Category	**Example attributes**
Financial preferences	Budget minimum and maximum
Academic details	University, course, year of study
Personal details	Age, nationality
Accommodation	Property type
Lifestyle preferences	Smoking, pet friendliness
Living habits	Cleanliness, noise tolerance

1.5 Preliminary Dataset Statistics

The initial dataset inspection identified no missing values and no exact duplicate rows. The following descriptive statistics summarize selected numerical attributes.

Attribute	Mean	Median	Minimum	Maximum
Compatibility score	69.20	71	10	99

The initial dataset inspection identified 1,333 records and 31 columns, with no missing values. The compatibility score ranges from 10 to 99, with a mean of approximately 69.20 and a median of 71. These statistics provide an overview of the compatibility scores in the dataset.

The dataset contains 710 records with scores greater than or equal to 70 and 623 records with scores below 70. This distribution provides two reasonably balanced classes for the binary classification experiment.

1.6 Target Variable and Compatibility Definition

The dataset contains an existing numerical target variable named compatibility_score. To convert the problem into a binary classification task, a threshold of 70 is applied.

The target variable is defined as follows:

- Compatibility score greater than or equal to 70: Compatible (1)
- Compatibility score below 70: Not Compatible (0)

This threshold is an experimental choice used to divide the compatibility scores into two classes.

The resulting dataset contains 710 Compatible records and 623 Not Compatible records.

The original numerical compatibility score is removed from the input features after creating the binary target to prevent target leakage. The remaining student attributes are used as input features for training the classification models.

2. Sampling and Experimental Methodology

2.1 Sampling Unit and Pair Formation

The sampling unit in this experiment is a student pair represented by one row in the supplied dataset. The dataset already contains pairwise information for two students along with their compatibility score. Therefore, additional pair generation from individual profiles is not required.

The dataset consists of 1,333 pair records. The compatibility score is converted into a binary target using a threshold of 70.

The dataset is divided into training and testing subsets using StratifiedGroupKFold with five folds. Approximately 80% of the records are used for training and 20% for testing.

2.2 Train-Test Sampling and Data Leakage

StratifiedGroupKFold is used to divide the dataset while maintaining class proportions and keeping repeated unordered student pairs together in the same partition.

A group identifier is constructed using the two student IDs in sorted order. This ensures that the same student pair is not present in both training and testing datasets, even if the order of the two students is reversed.

Both Logistic Regression and Random Forest are trained using the same training dataset and evaluated on the same held-out test dataset. This provides a fair basis for comparing their classification performance.

2.3 Data Preprocessing

1. Load the dataset using Pandas.
2. Inspect the dimensions, column names, data types, and missing values.
3. Separate the compatibility score from the input features.
4. Convert the compatibility score into a binary target using the threshold of 70.
5. Exclude student IDs, names, and free-text bio columns from predictive features.
6. Identify numerical and categorical features.
7. Apply median imputation and standard scaling to numerical features.
8. Apply most-frequent imputation and one-hot encoding to categorical features.

9. Fit preprocessing transformations using training data only to prevent data leakage.

2.4 Exploratory Data Analysis

Exploratory Data Analysis is performed to understand the characteristics of the supplied roommate compatibility dataset. The analysis includes examining numerical attributes, categorical preferences, compatibility score distributions, and class frequencies.

The compatibility score is analyzed to understand its range and distribution. Frequency distributions are also examined to understand the number of Compatible and Not Compatible records. These analyses help identify the characteristics of the dataset before model training.

2.5 Feature Selection and Preprocessing

Pairwise feature	**Example construction**	**Preprocessing**
Financial	Budget minimum and maximum	Numerical processing
Academic	University, course, year	Categorical encoding
Personal	Age, nationality	Numerical/categorical processing
Accommodation	Property type	Categorical encoding
Lifestyle	Smoking, pet friendliness	Categorical encoding
Living habits	Cleanliness, noise tolerance	Numerical/categorical processing

The original dataset already contains attributes for both students in each pair. Therefore, the experiment uses these existing paired attributes directly instead of generating additional pairwise similarity features. Student identifiers, names, and free-text bios are excluded from model training because they are not used as predictive features in the current implementation.

2.6 Compatibility Scoring Baseline

The compatibility score is converted into a binary classification target using a threshold of 70. Records with scores greater than or equal to 70 are assigned to the Compatible class, while records with scores below 70 are assigned to the Not Compatible class.

Two supervised machine learning algorithms are implemented for comparison. Logistic Regression is selected as the baseline classifier because it provides a simple and interpretable classification approach. Random Forest is selected as the candidate improved classifier because it combines multiple decision trees and can capture nonlinear relationships between input features.

Both models use the same preprocessed input features and the same training and testing partitions. Their performance is compared using classification metrics and statistical techniques.

2.7 Machine Learning Models

2.7.1 Logistic Regression

Logistic Regression is used as a baseline supervised classification algorithm. It estimates the probability of a pair belonging to the positive class using a logistic function applied to a linear combination of input features. In this project, the positive class is Compatible. A classification threshold is then used to assign the predicted class. Logistic Regression is relatively interpretable and provides a useful baseline for comparison.

2.7.2 Random Forest Classifier

Random Forest is an ensemble learning method that combines predictions from multiple decision trees. Each tree is trained using a sample of the training data and a subset of features. For classification, the trees vote for a class or contribute to an estimated class probability. Random Forest can model nonlinear relationships and interactions among pairwise lifestyle features. Its performance is evaluated using the same test observations as Logistic Regression.

2.7.3 Model Training and Comparison

Logistic Regression and Random Forest are trained using the same training dataset. Both models use identical input features and preprocessing procedures to ensure a fair comparison.

Logistic Regression serves as the baseline model, while Random Forest is evaluated as the candidate improved model. The trained models generate predictions for the same held-out test observations. Their performance is compared using accuracy, precision, recall, F1-score, and ROC-AUC.

2.8 Evaluation Metrics

The confusion matrix provides True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). The following measures are calculated with Compatible treated as the positive class.

Metric	**Formula**	**Meaning**
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Proportion of all test pairs classified correctly
Precision	TP / (TP + FP)	Among pairs predicted Compatible, proportion labelled Compatible
Recall	TP / (TP + FN)	Proportion of labelled Compatible pairs identified
Specificity	TN / (TN + FP)	Proportion of labelled Not Compatible pairs identified
F1-score	2 × Precision × Recall / (Precision + Recall)	Harmonic mean of precision and recall
ROC-AUC	$AUC = \int_0^1 TPR(FPR)d(FPR)$	distinguish between the positive and negative classes across different classification thresholds

3. Observations, Decision, and Limitations

3.1 Random Forest Confusion Matrix

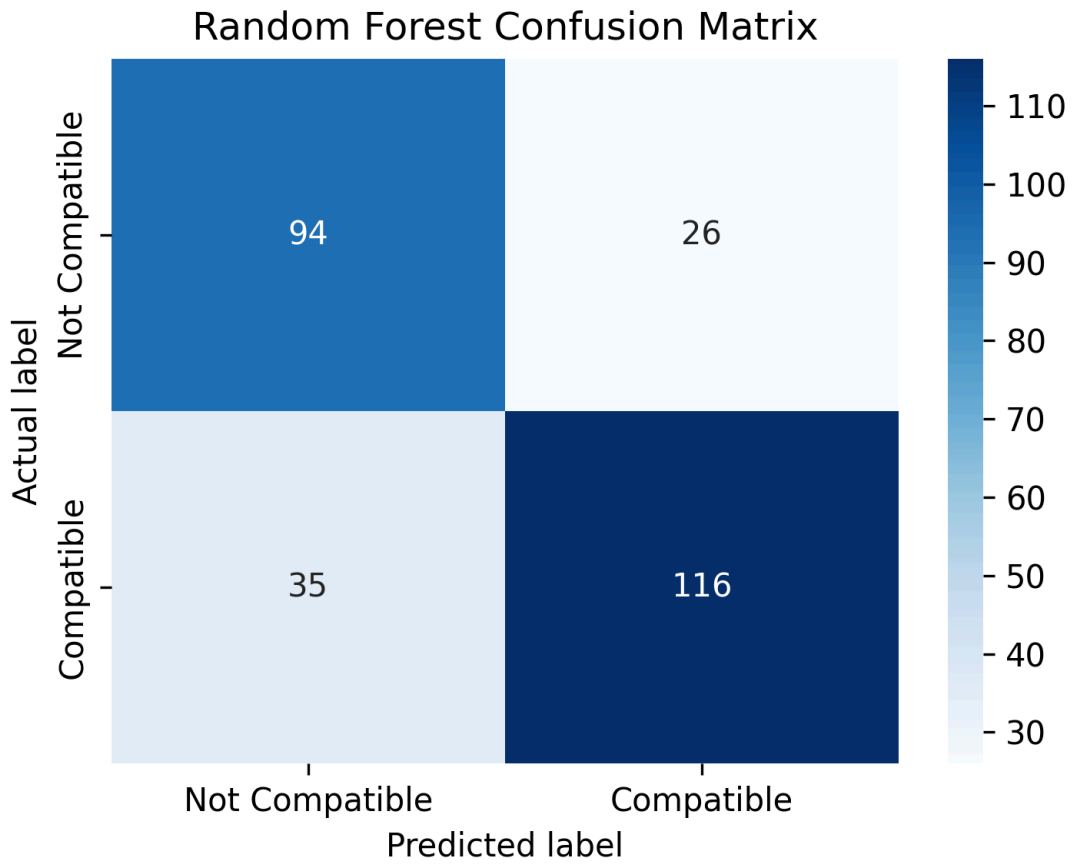


Figure 3.1: Confusion Matrix of the Random Forest Classifier

Actual / Predicted	**Not Compatible**	**Compatible**
Not Compatible	194	26
Compatible	35	116

The Random Forest classifier is evaluated on the held-out test dataset. Its confusion matrix provides the number of correctly and incorrectly classified observations for both compatibility classes.

The model achieved an accuracy of 71.16% on the test dataset. The confusion matrix helps identify the number of True Positives, True Negatives, False Positives, and False Negatives.

The results indicate that Random Forest provides a useful classification performance for predicting roommate compatibility based on the available student attributes.

3.2 Logistic Regression Confusion Matrix

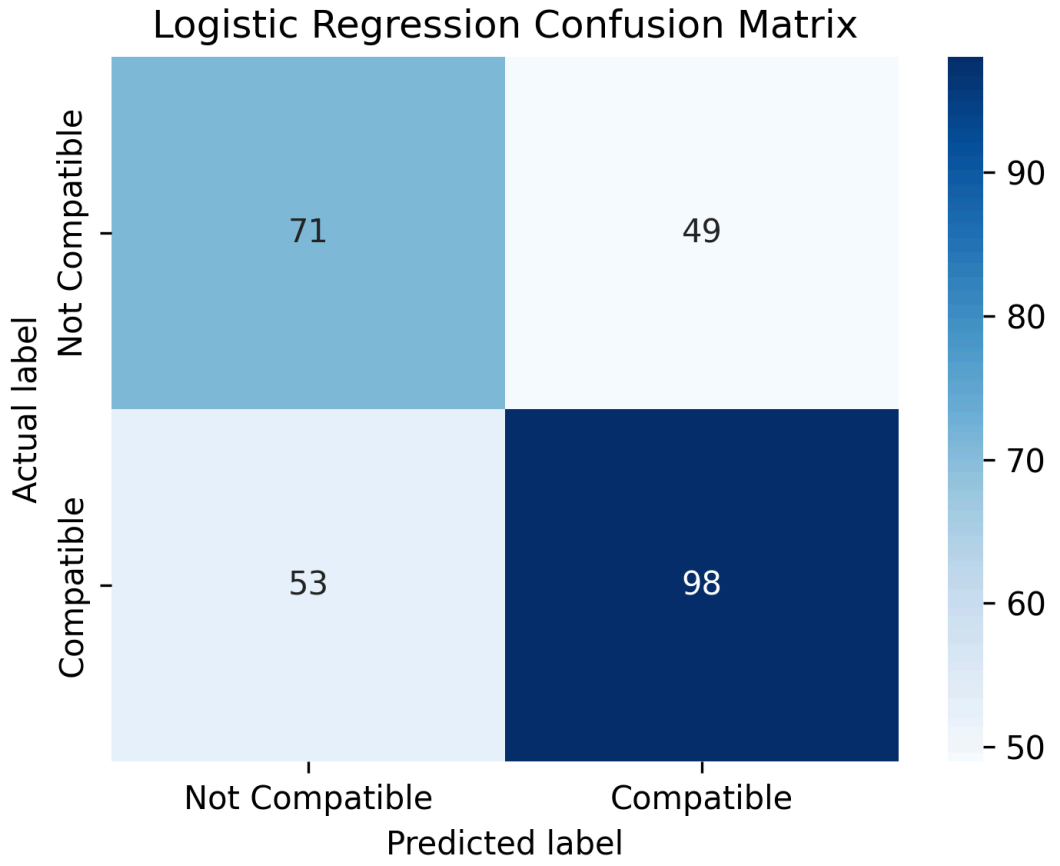


Figure 3.2: Confusion Matrix of the Logistic Regression Classifier

Actual / Predicted	**Not Compatible**	**Compatible**
Not Compatible	71	49
Compatible	53	98

Logistic Regression is used as the baseline classification model. Its performance is evaluated using the same held-out test observations used for the Random Forest classifier.

The model achieved an accuracy of 62.92%. The confusion matrix represents its classification outcomes for the Compatible and Not Compatible classes.

The results provide a baseline for evaluating whether the Random Forest classifier offers a meaningful improvement in predictive performance.

3.3 Comparison of Models

Evaluation metric	Logistic Regression	Random Forest
Accuracy	62.92%	71.16%
Precision (Compatible)	65.03%	75.19%
Recall (Compatible)	65.49%	68.31%
F1-score (Compatible)	65.26%	71.59%
ROC-AUC	65.25%	80.83%

3.4 Observations and Decision

- Both classifiers were evaluated using the same held-out test dataset.
- Logistic Regression achieved an accuracy of 62.92%.
- Random Forest achieved an accuracy of 71.16%.
- Random Forest demonstrated an improvement of 8.24 percentage points in accuracy.
- Random Forest also achieved higher precision, recall, F1-score, and ROC-AUC.
- Statistical testing is performed to determine whether the observed improvement is statistically significant.

3.5 Limitations

1. The compatibility labels are derived from an existing numerical compatibility score using an experimentally selected threshold of 70.
2. The classification results depend on the quality and consistency of the original compatibility scores.
3. The dataset may not represent the entire population of students seeking roommates.
4. Personal preferences are subjective, and a high predicted compatibility score does not guarantee successful cohabitation.
5. Repeated appearances of an individual in different student pairs may introduce statistical dependence.
6. Grouping identical unordered student pairs prevents duplicate-pair leakage, but does not guarantee that all individuals in the test set are unseen during training.

4. Application of Probability and Statistics

4.1 Descriptive Statistics

The compatibility score has a mean of approximately 69.20 and a median of 71. The minimum score is 10, while the maximum score is 99. These descriptive statistics summarize the distribution of compatibility scores in the supplied dataset.

The binary target contains 710 Compatible observations and 623 Not Compatible observations.

4.2 Frequency and Probability Distributions

Relative frequency is used to estimate the proportion of observations belonging to each compatibility class. For example, the proportion of Compatible observations is 710/1333, while the proportion of Not Compatible observations is 623/1333.

These empirical proportions describe the class distribution in the supplied dataset and help assess whether the classification problem is balanced.

4.3 Conditional Probability

Conditional probability measures the probability of an event given that another event has occurred:

$P(A \mid B) = P(A \cap B) / P(B)$, provided $P(B) > 0$

In this project, conditional probability can be used to examine questions such as the Probability of compatibility given similar budget ranges, or the Probability of compatibility given matching smoking preferences. These analyses reveal associations and should not be interpreted as causal effects.

4.4 Sampling and Sampling Probability

The dataset contains 1,333 student-pair observations. StratifiedGroupKFold with five folds is used to divide the dataset into training and testing subsets.

The grouping mechanism ensures that repeated unordered student pairs are assigned to the same partition. Stratification helps maintain the class distribution across the training and testing subsets.

Approximately 80% of the observations are allocated to training, while the remaining 20% are used for testing. Both classification models are evaluated on the same held-out test observations to ensure a fair comparison.

4.5 Correlation and Association Analysis

Correlation can be used to study associations between numerical attributes such as Student 1 age and Student 2 age and Student 1 budget and Student 2 budget, or Cleanliness preferences of both students. Pearson's correlation coefficient measures linear association:

$r = Cov(X,Y) / (\sigma X \sigma Y)$

The coefficient ranges from −1 to +1. Positive values indicate positive linear association, negative values indicate negative linear association, and values near zero indicate weak linear association. Spearman rank correlation may be more suitable for ordinal variables or monotonic relationships. Correlation does not imply causation.

4.6 Confusion Matrix as a Statistical Evaluation Tool

The confusion matrices summarize the observed outcomes of classification on the test set. Accuracy, precision, recall, specificity, and F1-score are calculated from the four confusion-matrix counts. Precision and recall are particularly relevant because a false-positive recommendation and a missed compatible candidate have different practical consequences.

Metric	Logistic Regression	Random Forest
True negatives	71	194
False positives	53	35
False negatives	49	26
True positives	98	116

4.7 Paired Bootstrap Analysis

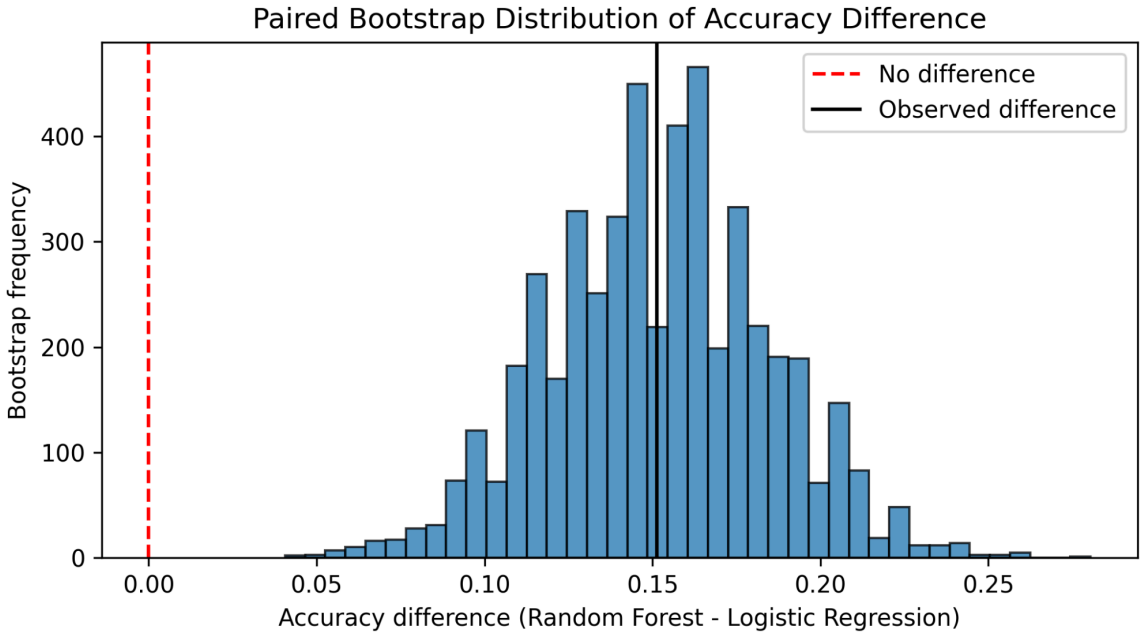


Figure 4.1: Paired Bootstrap Distribution of Accuracy Difference (Random Forest − Logistic Regression)

D_β = Accuracy(RF)$_\beta$ − Accuracy(LR)$_\beta$

Bootstrap sampling is a statistical resampling technique used to estimate the variability of a statistic. In this experiment, paired bootstrap sampling is used to estimate the uncertainty associated with the difference in accuracy between Logistic Regression and Random Forest.

Both models are evaluated on the same held-out test dataset. During each bootstrap iteration, test observations are sampled with replacement, and the accuracy difference between the two models is calculated using the same sampled observations.

The experiment performs 5,000 bootstrap iterations. The observed accuracy improvement of Random Forest over Logistic Regression is 8.24 percentage points.

The 95% bootstrap confidence interval ranges from 2.62% to 13.87%. Since the interval does not contain zero, the results provide evidence that the Random Forest classifier has higher accuracy than Logistic Regression on the evaluated test set.

The confidence interval also indicates the uncertainty associated with the estimated improvement. However, the interval is based on row-wise resampling and may not fully account for dependence between different pairs sharing an individual.

4.8 Hypothesis Testing Using McNemar's Test

Hypothesis testing is used to determine whether the observed improvement in model accuracy is statistically significant or could have occurred due to random variation in the test sample.

The exact one-sided McNemar test is applied because both classifiers are evaluated on the same test observations. The test compares the cases where one classifier is correct and the other classifier is incorrect.

Null Hypothesis (H_0):

The Random Forest classifier does not have a higher probability of correct classification than Logistic Regression.

Alternative Hypothesis (H_1):

The Random Forest classifier has a higher probability of correct classification than Logistic Regression.

The significance level is set to 0.05.

The exact one-sided McNemar test produces a p-value of 0.003574.

Since the p-value is less than 0.05, the null hypothesis is rejected. This provides statistical evidence that Random Forest performs better than Logistic Regression on the evaluated test dataset.

The result supports the observed improvement, although it does not establish that Random Forest will perform better on every future dataset.

5. Conclusion

This project investigated the application of machine learning and statistical techniques for evaluating roommate compatibility classification models. The experiment used the training_data.csv dataset

containing 1,333 student-pair records and 31 columns. The existing compatibility score was converted into a binary target using a threshold of 70.

Logistic Regression was selected as the baseline classifier, while Random Forest was evaluated as the improved classifier. Both models were trained and evaluated using the same held-out test dataset.

Random Forest achieved an accuracy of 71.16%, compared with 62.92% for Logistic Regression, representing an improvement of 8.24 percentage points. Random Forest also demonstrated higher precision, recall, F1-score, and ROC-AUC.

To investigate whether this improvement was statistically meaningful, paired bootstrap sampling was performed using 5,000 resamples. The resulting 95% confidence interval ranged from 2.62% to 13.87%. The exact one-sided McNemar test produced a p-value of 0.003574, which is below the significance level of 0.05. Therefore, the null hypothesis was rejected.

The results provide statistical evidence that Random Forest performs better than Logistic Regression on the evaluated test set. The experiment demonstrates the importance of probability and statistics in machine learning evaluation through sampling, estimation, confidence intervals, and hypothesis testing.

However, the findings are limited by the quality of the supplied compatibility scores, the experimental classification threshold, and the possibility of dependence between pairs sharing individual students. Further validation using independently collected real-world compatibility outcomes and profile-level evaluation would help establish the generalizability of the results.